

SUMMARY

Current network verification researcher and former enterprise-scale software engineer. Proficient in both application- and infrastructure-level development with expertise in SWE practices, including Agile, DevOps, IaaS, CI/CD. Experienced in R&D of high-performance, low-latency systems. Strong background in mathematics, algorithms, and formal verification. Passionate about architecting **maintainable**, **reliable**, and **large-scale** software systems.

EDUCATION

2024 - Present Ph.D. in Computer Science, Cheriton School of CS, University of Waterloo
2019 - 2022 M.Sc. in Software Engineering, School of ECE, University of Tehran
2014 - 2019 B.Sc. in Computer Engineering, School of ECE, University of Tehran

PUBLICATION

NSDI'26 Seyhani A, Gupta A, Walker D, Arashloo MT. *“Count-Based Abstractions for Performance Verification of Contention Points”*. 23rd USENIX Symposium on Networked Systems Design and Implementation.
HotNets'24 Seyhani A, Zhao J, Gupta A, Walker D, Arashloo MT. *“Buffy: A Formal Language-Based Framework for Network Performance Analysis”*. 23rd ACM Workshop on Hot Topics in Networks.

WORK EXPERIENCE

Software Engineer → Senior SWE, [Radin Bourse](#) (Tehran, Iran) (part-time) 2018 - 2023

- Designed and developed an ultra-low-latency trading engine for Tehran stock exchange using LMAX Disruptor. Achieving $\sim 50k$ TPS with $20 \mu s$ latency for sequential order processing.
- Led design and migration of an on-premise/bare-metal stack to an IaaS solution using Ansible, MaaS, Docker, Terraform, Nomad, Vault, and Consul.

Backend Developer, [Tapsi](#) (Tehran, Iran) (part-time) 2017 - 2018

Developed and deployed backend features for Tapsi, the second-largest online taxi platform in Iran.

Fullstack Developer, [chibaki.ir](#) (Tehran, Iran) (part-time) 2016 - 2017

Independently designed, developed, and deployed the backend of an online platform for connecting clients to professionals, using Node.js.

NOTABLE PROJECTS

[Count Buffy](#) (2026) Verification framework built on Z3 for checking performance properties of network schedulers, supporting realistic buffer sizes while having efficient verification time.

[Estrest](#) (2022) A tool for checking actual causality using event-structure semantics of programs.

[APHTTP](#) (2017) A light-weight Express.js inspired HTTP web-server and template rendering implemented with pure C++, to enable simple web development with C++.

SKILLS

Languages	Java, C++, Python, C, SQL, Javascript
Frameworks & Libraries	Spring Frameworks (Boot, Data, Security), JMH, Lombok, Cucumber, Hibernate, Z3, Node.js, NumPy, Pandas, Scikit-learn
Tools & Technologies	Ansible, Hazelcast, Maven, Terraform, Packer, Vault, Nomad, Consul, Docker, MySQL, MongoDB, JFR, GitLab CI, NGINX, mCRL2, Linux, Git, AWS, GCP, Prometheus, Grafana
Additional Experience	Kafka, OpenCV, RabbitMQ, ELK Stack